

Lecture 14

The quantum inverse scattering method, and its classical limit

Objectives. Course map: G2 and G5, P2 quantised. [Lecture 10](#) solved the Heisenberg chain by a guess and left the question of why it is solvable. This lecture answers with [Lecture 9](#)'s three moves, quantised. The chain's Lax operator satisfies an RLL relation with Yang's R -matrix of [Lecture 13](#), the monodromy satisfies RTT, and the transfer matrices commute at all values of the spectral parameter. Then the lecture restores $\hbar$ in the quantum Toda chain and recovers [Lecture 9](#)'s r -matrix as $\hbar \rightarrow 0$, sign for sign.

14.1 The question

[Lecture 9](#) made the Toda ring's integrals commute with one identity for one site; [Lecture 10](#) solved the Heisenberg chain by a guess. So the question is

why do the Heisenberg chain's charges commute, and what classical structure is this the quantisation of?

The answer: [Lecture 9](#)'s structure, with commutators for brackets and Yang's R for r ([Theorems 14.1 to 14.3](#)); and the classical limit of the quantum Toda chain proves that the dictionary is right ([Corollary 14.6](#)). The worked example is the Heisenberg chain's Lax operator.

14.2 From brackets to commutators

[Lecture 9](#)'s site matrices obey the quadratic bracket (9.9); it composes along an ultralocal chain ([Theorem 9.7](#)), and its trace makes the transfer matrices Poisson-commute ([Corollary 9.8](#)). Quantisation makes the entries of L operators and brackets commutators, $\{\cdot, \cdot\} \rightarrow [\cdot, \cdot] / i\hbar$, and the legs of [Definition 9.2](#) survive. The quantum relation for the monodromy is the RTT relation

$$R_{12}(\lambda - \mu) T_2(\mu) T_1(\lambda) = T_1(\lambda) T_2(\mu) R_{12}(\lambda - \mu), \quad R = \mathbb{1} + i\hbar r + O(\hbar^2). \quad (14.1)$$

At order $\hbar$ it returns (9.9); with the other order of the monodromies it would return $r \rightarrow -r$, so the order is forced (§ 14.6). Other books use the other order, each with its own R and Lax operator: convert all three together, never one alone. Until § 14.6, $\hbar = 1$, and the auxiliary copies of $\mathbb{C}^2$ are a and b , apart from the sites $n = 1, \dots, N$.

14.3 The Lax operator of the Heisenberg chain

The quantum space is the Heisenberg chain of [Lecture 10](#), $\mathcal{H} = (\mathbb{C}^2)^{\otimes N}$, one spin $\frac{1}{2}$ per site. The auxiliary space a is one more $\mathbb{C}^2$. The *Lax operator* at site n is

$$L_{an}(\lambda) = \lambda + iP_{an} = \left(\lambda + \frac{i}{2}\right) + i\sigma_a \cdot \mathbf{S}_n = \begin{pmatrix} \lambda + \frac{i}{2} + iS_n^z & iS_n^- \\ iS_n^+ & \lambda + \frac{i}{2} - iS_n^z \end{pmatrix}_a, \quad (14.2)$$

with P_{an} the permutation of the auxiliary space with site n ([Definition 9.4](#)), $\mathbf{S} = \sigma/2$ and $S^\pm = S^x \pm iS^y$. The middle form uses $P = \frac{1}{2}(\mathbb{1} + \sigma_a \cdot \sigma_n)$, the identity behind the Hamiltonian (10.2); the matrix is in the auxiliary space, with entries operators on site n . Like the Toda site matrix it is linear in the spectral parameter and acts on one site.

[Figure 14.1](#) draws it. Each space is a line: the auxiliary space horizontal, the site vertical. An operator on both is a vertex where they cross, and (14.2) says what happens there: with weight λ the two states go straight through, and with weight i they swap lines, the P of [Definition 9.4](#).

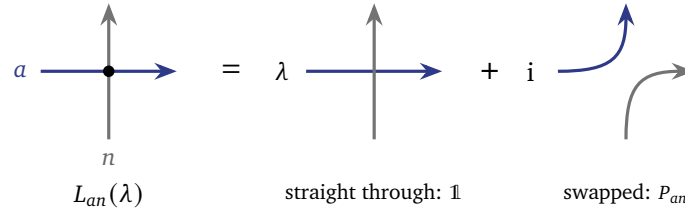


Figure 14.1: The Lax operator as a vertex. The auxiliary line (indigo) crosses the line of site n (grey). With weight λ both states go straight on; with weight i the auxiliary state leaves on the site's line and the site's state on the auxiliary line. (Where lines run side by side, a swap is drawn as a crossing; for lines that meet at right angles the crossing is the identity and the swap is two turns.)

Two auxiliary spaces a, b and one site n make three copies of $\mathbb{C}^2$, and on them the only operators needed are the three transpositions P_{ab}, P_{an}, P_{bn} . Their products are permutations of three objects, and a product of two different transpositions is a 3-cycle, of which there are two:

$$C = P_{ab}P_{an} = P_{an}P_{bn} = P_{bn}P_{ab}, \quad C^{-1} = P_{an}P_{ab} = P_{ab}P_{bn} = P_{bn}P_{an}. \quad (14.3)$$

(Follow a vector $x \otimes y \otimes z$ through each product; this is the computation of [Proposition 13.4](#).)

Theorem 14.1 (The RLL relation). With $R(\lambda) = \lambda - iP$,

$$R_{ab}(\lambda - \mu) L_{bn}(\mu) L_{an}(\lambda) = L_{an}(\lambda) L_{bn}(\mu) R_{ab}(\lambda - \mu) \quad (14.4)$$

for all λ, μ . With $\lambda + iP$ in place of R it fails.

Proof. Take $R = (\lambda - \mu) + siP_{ab}$ with a sign $s = \pm 1$ and compute the difference Δ of the two sides. It has two parts. The first is $(\lambda - \mu)(L_{bn}L_{an} - L_{an}L_{bn})$, and in $L_{bn}L_{an} - L_{an}L_{bn}$ everything cancels except the products of two P 's: $P_{an}P_{bn} - P_{bn}P_{an} = C - C^{-1}$. The second is

$$si(P_{ab}L_{bn}L_{an} - L_{an}L_{bn}P_{ab}) = si\left(i\mu(P_{ab}P_{an} - P_{an}P_{ab}) + i\lambda(P_{ab}P_{bn} - P_{bn}P_{ab}) - (P_{ab}P_{bn}P_{an} - P_{an}P_{bn}P_{ab})\right),$$

the terms $\lambda\mu P_{ab}$ having cancelled. By (14.3) the first bracket is $C - C^{-1}$, the second $C^{-1} - C$, and the triple products are both P_{bn} , so they cancel. The second part is therefore $s(\lambda - \mu)(C - C^{-1})$, and

$$\Delta = (1 + s)(\lambda - \mu)(C - C^{-1}).$$

This vanishes for $s = -1$ and is not zero for $s = +1$, since $C \neq C^{-1}$. ■

The whole relation is a statement about the permutations of three objects (the check script confirms the residual symbolically).

14.4 From one site to the chain

The monodromy matrix

The *monodromy matrix* of the chain is the ordered product of the Lax operators along it,

$$T_a(\lambda) = L_{aN}(\lambda) \cdots L_{a2}(\lambda) L_{a1}(\lambda) = \begin{pmatrix} A(\lambda) & B(\lambda) \\ C(\lambda) & D(\lambda) \end{pmatrix}_a, \quad (14.5)$$

an operator on $\mathbb{C}_a^2 \otimes \mathcal{H}$: in the auxiliary space a 2×2 matrix whose entries A, B, C, D are operators on the whole chain. As a picture it is a row: one auxiliary line crossing every site in turn (Figure 14.2), Figure 9.1 with operators at the vertices. Operators on different sites commute, so L_{an} commutes with L_{bm} for $n \neq m$: they act on disjoint sets of spaces. That is the quantum form of ultralocality, and it is all the next proof uses.

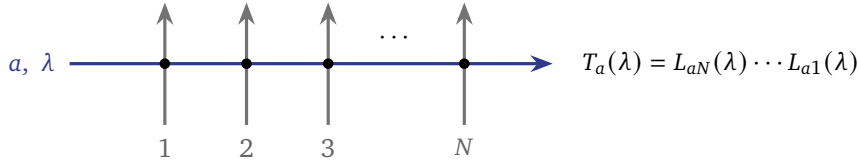


Figure 14.2: The monodromy matrix as a row. The auxiliary line meets site 1 first, so L_{a1} is the rightmost factor. The two free ends of the auxiliary line are the two indices of the 2×2 matrix (14.5); the ends of the site lines are the quantum space.

Theorem 14.2 (RLL implies RTT). *The monodromy (14.5) satisfies*

$$R_{ab}(\lambda - \mu) T_b(\mu) T_a(\lambda) = T_a(\lambda) T_b(\mu) R_{ab}(\lambda - \mu). \quad (14.6)$$

The same holds for any chain whose Lax operators obey (14.4) with a common R and act on different sites.

Proof. In $T_b T_a = L_{bN} \cdots L_{b1} L_{aN} \cdots L_{a1}$, move each L_{an} to the left past $L_{b,n-1}, \dots, L_{b1}$, with which it commutes. The factors pair up by site:

$$T_b(\mu) T_a(\lambda) = (L_{bN} L_{aN}) (L_{b,N-1} L_{a,N-1}) \cdots (L_{b1} L_{a1}).$$

Now R_{ab} enters on the left. By (14.4) at site N , $R_{ab} L_{bN} L_{aN} = L_{aN} L_{bN} R_{ab}$: it has passed the first pair and reversed it. The same step at each site in turn carries R_{ab} to the right end, leaving $(L_{aN} L_{bN}) \cdots (L_{a1} L_{b1}) R_{ab}$, and undoing the pairing gives $T_a T_b R_{ab}$. ■

This is the *train argument* (Figure 14.3): the crossing of the two auxiliary lines is pushed through the row one site at a time, as the bracket was pushed through the product by Leibniz's rule in Theorem 9.7.

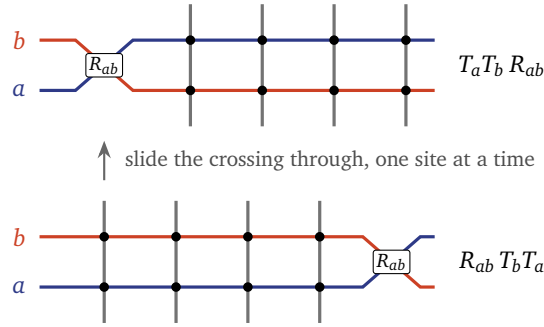


Figure 14.3: The train argument. Operators act left to right along the auxiliary lines, so the leftmost thing drawn is the rightmost factor. Bottom: the rows, then the crossing R_{ab} . By (14.4) the crossing slides through site N , then $N - 1$, $\dots$, until it has passed the row (top): (14.6).

The Lax operator is an R -matrix, and RLL is the Yang–Baxter equation

Compare (14.2) with R :

$$L_{an}(\lambda) = \lambda + iP_{an} = -R_{an}(-\lambda). \quad (14.7)$$

The Lax operator is the R -matrix itself, with the site as its second space. It is *not* $R_{an}(\lambda)$: the two differ by $\lambda \rightarrow -\lambda$. Substitute (14.7) into (14.4); the two minus signs cancel,

$$R_{ab}(\lambda - \mu) R_{bn}(-\mu) R_{an}(-\lambda) = R_{an}(-\lambda) R_{bn}(-\mu) R_{ab}(\lambda - \mu).$$

Rename $b \rightarrow 1$, $a \rightarrow 2$, $n \rightarrow 3$, and put $u = -\mu$, $v = -\lambda$, so that $\lambda - \mu = u - v$. Since $P_{12} = P_{21}$ this is

$$R_{12}(u - v) R_{13}(u) R_{23}(v) = R_{23}(v) R_{13}(u) R_{12}(u - v), \quad (14.8)$$

the *Yang–Baxter equation* in its standard form. With $u = u_{13}$, $v = u_{23}$ and $u - v = u_{12}$ it is the equation (13.6) that [Theorem 13.3](#) derived from three-body consistency, for the matrix $X_B(u) = R(u)/(u + i)$ of the delta bosons (13.3). So [Theorem 14.1](#) and [Proposition 13.4](#) are one fact, read twice. In [Lecture 13](#) the three lines were three particles and R a two-body collision; in the Heisenberg chain one of the three lines is a site. The fermions of [Lecture 13](#) scatter with $X_F = -L(u)/(u + i)$, the Lax operator itself.

[Figure 14.5](#) shows the equation as a picture. It is [Figure 13.1](#), and [Figure 14.4](#) is its special case with one line vertical.

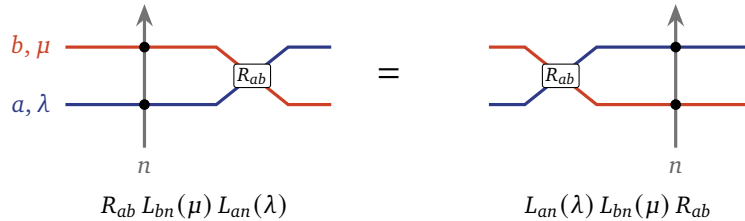


Figure 14.4: The *RLL* relation (14.4). Left: the site line n meets a (lower) and then b , and afterwards the auxiliary lines cross. Right: they cross first, and n meets b and then a . The crossing of a and b may be slid through the site line.

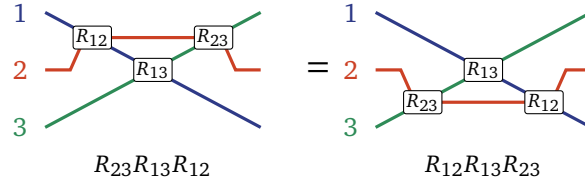


Figure 14.5: The Yang–Baxter equation (14.8) as a slide: line 2 passes either above or below the crossing of lines 1 and 3. Lines carry spectral parameters and a crossing of lines i, j carries R_{ij} at the difference of their parameters, which is why $u_{13} = u_{12} + u_{23}$. Reading left to right, the first crossing is the rightmost factor. Make line 3 a site and this is Figure 14.4.

14.5 Commuting transfer matrices

The *transfer matrix* is the trace of the monodromy over the auxiliary space,

$$t(\lambda) = \text{tr}_a T_a(\lambda) = A(\lambda) + D(\lambda), \quad (14.9)$$

an operator on the chain alone: in the picture the auxiliary line is closed into a loop (Figure 14.6). It is a polynomial of degree N in λ with operator coefficients, and leading term $2\lambda^N$.

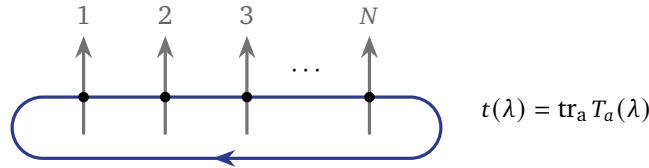


Figure 14.6: The transfer matrix: the auxiliary line of Figure 14.2, closed into a loop by the trace. The loop has no ends, so it acts on the sites alone; this is the loop of Figure 9.1 with operators at the vertices.

Theorem 14.3 (Commuting transfer matrices). $[t(\lambda), t(\mu)] = 0$ for all $\lambda, \mu \in \mathbb{C}$.

Proof. $R(u)R(-u) = (u - iP)(-u - iP) = -(u^2 + 1)$, so $R(u)$ is invertible for $u \neq \pm i$. For such $\lambda - \mu$, multiply (14.6) on the left by R_{ab}^{-1} : $T_b(\mu)T_a(\lambda) = R_{ab}^{-1}T_a(\lambda)T_b(\mu)R_{ab}$. Take the trace over both auxiliary spaces. On the left, $\text{tr}_{ab} T_b T_a = t(\mu)t(\lambda)$, because T_a and T_b have their auxiliary legs in different spaces. On the right, R_{ab} acts on the auxiliary spaces only, as a numerical matrix, and the partial trace is cyclic for such a factor; so $\text{tr}_{ab} (R_{ab}^{-1}T_a T_b R_{ab}) = \text{tr}_{ab} (T_a T_b) = t(\lambda)t(\mu)$. Both sides are polynomials in λ, μ , so the identity extends to $\lambda - \mu = \pm i$. ■

This is Corollary 9.8: there the trace of a commutator vanished, here the trace of a conjugate is the trace (Figure 14.7). The coefficients of $t(\lambda)$ commute with one another. Whether the family contains a Hamiltonian worth having, and whether its members are local, is the business of Lecture 15.

What *RTT* gives, and what it does not

The proof used (14.6) and the invertibility of R , and not the Yang–Baxter equation. The equation does two other jobs. It *supplies Lax operators*: by (14.7) the rational R , read with

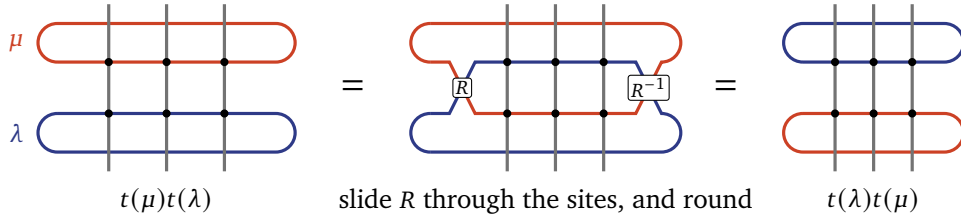


Figure 14.7: Why transfer matrices commute. Insert $RR^{-1} = \mathbb{1}$ between the loops; R slides through every site (Figure 14.3), exchanging the rows, and meets R^{-1} round the back. The loops have passed through each other.

a site as its second space, is a Lax operator for its own RLL relation, which is why the Heisenberg chain has one. And it is *sufficient, not forced*, for the consistency of the RTT relations taken as the definition of an algebra (the readings at the end). Finally, *integrability does not require RTT* : the free-fermion charges of the Ising chain (Lecture 18) are constructed directly.

14.6 The classical limit, verified: the quantum Toda chain

The Heisenberg chain has no classical limit worth the name — a spin $\frac{1}{2}$ has no small parameter. To test the correspondence of § 14.2, quantise Lecture 9's chain. Restore $\hbar$. The periodic Toda chain has canonical pairs with $[q_m, p_n] = i\hbar \delta_{mn}$, represented by $p_n = -i\hbar \partial/\partial q_n$ on functions of q . Keep the site matrix of (9.2) as it stands,

$$L_n(\lambda) = \begin{pmatrix} \lambda - p_n & e^{-q_n} \\ -e^{q_n} & 0 \end{pmatrix}, \quad (14.10)$$

now with operator entries. No ordering question arises: each entry involves q_n or p_n , never both.

Theorem 14.4 (Quantum Toda: RLL). *The site matrix (14.10) satisfies*

$$R_{ab}(\lambda - \mu) L_b(\mu) L_a(\lambda) = L_a(\lambda) L_b(\mu) R_{ab}(\lambda - \mu), \quad R(\lambda) = \lambda - i\hbar P,$$

exactly, to all orders in $\hbar$, and fails with $\lambda + i\hbar P$. Consequently the monodromy $T = L_N \cdots L_1$ satisfies RTT , and $[t(\lambda), t(\mu)] = 0$.

Proof. Write $L(\lambda) = \lambda E_{11} + L^0$ with $L^0 = L(0)$. In the leg notation the $(ij), (kl)$ entry of $L_a(\lambda)L_b(\mu)$ is $L_{ik}(\lambda)L_{jl}(\mu)$, and of $L_b(\mu)L_a(\lambda)$ it is the same product in the other order. So their difference is the matrix of commutators $-[L_{ik}, L_{jl}]$, which do not depend on λ, μ . The only non-zero ones are $[p, e^{\mp q}] = \pm i\hbar e^{\mp q}$, exactly $i\hbar$ times the Poisson brackets used in the proof of Proposition 9.9; that proof computed the whole matrix of brackets, so

$$L_b(\mu)L_a(\lambda) - L_a(\lambda)L_b(\mu) = -i\hbar (E_{11} \otimes L^0 - L^0 \otimes E_{11})P.$$

Next, $P L_b(\mu)L_a(\lambda)P = L_a(\mu)L_b(\lambda)$ (conjugating by P exchanges the legs), and expanding $L = \lambda E_{11} + L^0$ gives $L_a(\mu)L_b(\lambda) - L_a(\lambda)L_b(\mu) = (\mu - \lambda)(E_{11} \otimes L^0 - L^0 \otimes E_{11})$; the $L^0 \otimes L^0$ terms, in the same order on both sides, cancel. With $R = (\lambda - \mu) + si\hbar P$ the difference of the two sides of the relation is

$$(\lambda - \mu)(L_b L_a - L_a L_b) + si\hbar (P L_b L_a - L_a L_b P) = -(1 + s) i\hbar (\lambda - \mu)(E_{11} \otimes L^0 - L^0 \otimes E_{11})P,$$

using $PL_bL_a = L_a(\mu)L_b(\lambda)P$. It vanishes for $s = -1$ only. Sites commute, so [Theorem 14.2](#) and [Theorem 14.3](#) apply verbatim, now with $R(u)R(-u) = -(u^2 + \hbar^2)$. ■

The coefficients of $t(\lambda) = \sum_k c_k \lambda^{N-k}$ are the quantum integrals. They are the expressions of (9.4), now as operators, with no ordering ambiguity since each term has at most one factor per site: $c_1 = -\sum p_n$, and $H = \frac{1}{2}c_1^2 - c_2 = \frac{1}{2}\sum p_n^2 + \sum e^{q_n - q_{n+1}}$ is the quantum periodic Toda Hamiltonian, a function of the commuting c_k . Its spectral problem was reduced to one-variable quantisation conditions, for small N by Gutzwiller (1981) and for all N by Sklyanin's separation of variables (1985) · *cited*.

$\hbar \rightarrow 0$

To compare with [Lecture 9](#), divide by $\lambda - \mu$:

$$\frac{R(\lambda - \mu)}{\lambda - \mu} = \mathbb{1} + i\hbar r(\lambda - \mu), \quad r(\lambda) = -\frac{P}{\lambda},$$

exactly the r of [Proposition 9.9](#), and (14.1)'s expansion with no $O(\hbar^2)$ term. The limit needs a rule for turning operators into functions.

Lemma 14.5 (Symbols). Write an operator polynomial in the p 's with all p 's on the right, $F = \sum_\alpha f_\alpha(q) p^\alpha$, and call $\sigma(F) = \sum_\alpha f_\alpha(q) p^\alpha$, a function of (q, p) , its symbol. Then

$$\sigma(FG) = \sum_\alpha \frac{(-i\hbar)^{|\alpha|}}{\alpha!} \partial_p^\alpha \sigma(F) \partial_q^\alpha \sigma(G), \quad \text{so} \quad \sigma([F, G]) = i\hbar \{\sigma(F), \sigma(G)\} + O(\hbar^2).$$

Proof (★). Move each $p_n = -i\hbar \partial_{q_n}$ in F to the right through G 's functions of q by Leibniz's rule, $p_n^k g = \sum_j \binom{k}{j} (-i\hbar)^j (\partial_{q_n}^j g) p_n^{k-j}$. The first-order terms of $\sigma(FG) - \sigma(GF)$ are $-i\hbar \sum_n (\partial_{p_n} f \partial_{q_n} g - \partial_{p_n} g \partial_{q_n} f) = i\hbar \{f, g\}$ in the course's convention. ■

Corollary 14.6 (The classical limit of RTT). The symbols of the entries of the quantum monodromy are the entries of the classical monodromy (9.3). At order $\hbar$ the RTT relation of [Theorem 14.4](#) becomes

$$\{T_1(\lambda), T_2(\mu)\} = [r(\lambda - \mu), T_1(\lambda)T_2(\mu)], \quad r(\lambda) = -\frac{P}{\lambda} :$$

[Proposition 9.9](#) and [Theorem 9.7](#), sign for sign.

Proof. Each entry of T is a sum of products of entries of different site matrices, which commute, so its symbol is the classical product. Write RTT as $T_a T_b - T_b T_a = i\hbar(r T_b T_a - T_a T_b r)$ and take symbols. On the left the $(ij), (kl)$ entry is the commutator $[T_{ik}(\lambda), T_{jl}(\mu)]$, whose symbol is $i\hbar \{T_{ik}, T_{jl}\} + O(\hbar^2)$ by [Lemma 14.5](#): the entry of $i\hbar \{T_1, T_2\}$ in [Definition 9.2](#). On the right the symbols of $T_b T_a$ and $T_a T_b$ are both $T_1 T_2 + O(\hbar)$. Divide by $i\hbar$ and let $\hbar \rightarrow 0$. ■

Tracing over both legs, $\{t(\lambda), t(\mu)\} = 0$ is the order- $\hbar$ term of [Theorem 14.3](#). With $\lambda + i\hbar P$, false for this chain, the same steps would give $r = +P/\lambda$: the classical side decides the quantum sign. The check script repeats this computation for two and three sites.

14.7 What it buys

- *A commuting family for the Heisenberg chain*, from one relation for one site ([Theorems 14.1 to 14.3](#)). [Lecture 15](#) obtains the chain's Hamiltonian from its logarithmic derivative and diagonalises the family.
- *The Yang–Baxter equation, read a second time*: the Lax operator is Yang's R with a site as one of its spaces, and RLL is the equation of [Lecture 13](#) (14.8).
- *G5, closed*: the quantum Toda chain's RTT relation becomes [Lecture 9](#)'s quadratic bracket as $\hbar \rightarrow 0$, sign for sign ([Corollary 14.6](#)).

14.8 Caveats, in one place

- *Status*. [Theorems 14.1 to 14.4](#) and [Corollary 14.6](#) are proved; the proof of [Lemma 14.5](#) is a reading. The quantum Toda chain's spectral problem is cited.
- *What RTT does not give*. It gives commuting operators, not a Hamiltonian worth having or locality ([Lecture 15](#)), and not completeness, which stays [Lecture 10](#)'s cited question. The independence of the quantum Toda integrals is not shown.
- *The classical limit is tested on Toda*; the Heisenberg chain has no small parameter.

Tutorial

14.9 Two sites by hand

For $N = 2$, $t(\lambda) = \text{tr}_a [(\lambda + iP_{a2})(\lambda + iP_{a1})] = 2\lambda^2 + 2i\lambda - \text{tr}_a (P_{a2}P_{a1})$, using $\text{tr}_a \mathbb{1} = 2$ and $\text{tr}_a P_{an} = 1$. By (14.3), $P_{a2}P_{a1} = P_{21}P_{a2}$, so $\text{tr}_a (P_{a2}P_{a1}) = P_{12}$ and

$$t(\lambda) = 2\lambda^2 + 2i\lambda - P_{12},$$

which commutes with itself at any two values, as it must: it involves only P_{12} . On the triplet $P_{12} = 1$, on the singlet -1 . Then $t(0) = -P_{12}$, the swap of the two sites, and $t(0)^{-1}t'(0) = (-P_{12})(2i) = -2iP_{12}$: [Lecture 15](#) finds the chain's Hamiltonian in this logarithmic derivative. Repeat for $N = 3$ and check that $t(0) = -iP_{32}P_{31}$.

Reading

14.10 The classical Yang–Baxter equation

The linear Sklyanin bracket (9.11) of [Lecture 9](#)'s readings is an assumption about a Poisson bracket, so it must be consistent with the Jacobi identity. Working out what that requires produces the classical form of [Lecture 13](#)'s equation.

Put three legs in play, $V \otimes V \otimes V$, write L_i for L in leg i at spectral parameter λ_i , and abbreviate $r_{ij} = r_{ij}(\lambda_i - \lambda_j)$. Two facts do the work: $[r_{ij}, L_k] = 0$ when $k \notin \{i, j\}$, because they act on disjoint legs; and $r_{21} = -r_{12}$, which [Remark 9.13](#) showed to be *sufficient* for antisymmetry of the Poisson bracket but which the bracket does not by itself force. It is therefore a hypothesis of [Theorem 14.7](#), not a consequence of [Definition 9.12](#), and is assumed from here on.

Theorem 14.7 (Jacobi and the CYBE). For a Lax matrix satisfying (9.11) with an r obeying the skew condition (9.12), the Jacobi identity for L_1, L_2, L_3 is equivalent to

$$[[r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}], L_1 + L_2 + L_3] = 0. \quad (14.11)$$

In particular it holds whenever r satisfies the classical Yang–Baxter equation

$$[r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}] = 0. \quad (14.12)$$

Proof. Since r is numerical it passes through a Poisson bracket, so

$$\{\{L_1, L_2\}, L_3\} = [r_{12}, \{L_1, L_3\} + \{L_2, L_3\}] = [r_{12}, [r_{13}, L_1 + L_3] + [r_{23}, L_2 + L_3]].$$

Use $[r_{12}, L_3] = 0$ and the *matrix* Jacobi identity in the form $[A, [B, C]] = [[A, B], C] + [B, [A, C]]$ on each of the two terms. The first gives

$$[r_{12}, [r_{13}, L_1 + L_3]] = [[r_{12}, r_{13}], L_1 + L_3] + [r_{13}, [r_{12}, L_1]],$$

and the second the same with (13) $\rightarrow$ (23), $L_1 \rightarrow L_2$. Now add the two cyclic permutations of this, replacing r_{21}, r_{31}, r_{32} by $-r_{12}, -r_{13}, -r_{23}$ as (9.12) allows. The six leftover double commutators pair up, again by matrix Jacobi: for instance

$$[r_{13}, [r_{12}, L_1]] - [r_{12}, [r_{13}, L_1]] = -[[r_{12}, r_{13}], L_1].$$

Now collect. The first group contributes each double commutator of r 's twice, once from each of the two cyclic terms that produce it, and the two contributions carry different pairs of L_i :

$$[r_{12}, r_{13}] : L_1 + L_2 \text{ and } L_1 + L_3; \quad [r_{12}, r_{23}] : L_1 + L_2 \text{ and } L_2 + L_3; \quad [r_{13}, r_{23}] : L_1 + L_3 \text{ and } L_2 + L_3,$$

so their totals are $2L_1 + L_2 + L_3$, $L_1 + 2L_2 + L_3$ and $L_1 + L_2 + 2L_3$ respectively. The three surviving pairs of double commutators, computed as displayed above, contribute exactly $-[[r_{12}, r_{13}], L_1]$, $-[[r_{12}, r_{23}], L_2]$ and $-[[r_{13}, r_{23}], L_3]$, which remove the one doubled term in each line. All three are left with $L_1 + L_2 + L_3$, and that is (14.11). ■

Remark 14.8 (Machine-checked). The collection above is exactly the kind of bookkeeping in which a sign is lost. It is verified in `scripts/check-l8-algebra.py`, for $d = 2$, with three explicit legs and *symbolic* entries throughout — sixty free parameters, so the result is an identity and not a sample. The Jacobi sum equals $[X, L_1 + L_2 + L_3]$, and — the point of Remark 14.9 — is *not* identically zero on its own.

Remark 14.9 (Jacobi does not force the CYBE, and saying so matters). Theorem 14.7 gives (14.11), and (14.12) is *sufficient* for it, not implied by it: the Jacobi identity only requires the triple sum to *commute with* $L_1 + L_2 + L_3$, and for a particular L that is a weaker demand. What makes (14.12) the right thing to impose is that it mentions no L at all. One solution r then serves every model whose Lax matrix it fits, which is why the classification of r -matrices is a subject and the classification of integrable models is not.

One variant deserves its name in passing. Replacing the zero on the right of (14.12) by a specific r -independent term gives the *modified classical Yang–Baxter equation*, whose solutions generate the same structures over a slightly larger class of Lie algebras. Its theory belongs to Lie-bialgebra and Poisson–Lie theory, which this course does not enter; it is named here so that meeting the phrase elsewhere is not a surprise.

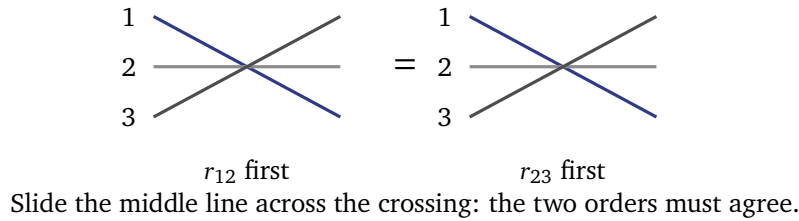


Figure 14.8: The classical Yang–Baxter equation. Three lines cross pairwise, and the two ways of resolving the triple crossing — sliding line 2 past the crossing of 1 and 3 either before or after — give the same answer. This is the classical form of Figure 13.1.

Reading

The rational solution

Proposition 14.10 (The rational r -matrix). $r(\lambda) = cP/\lambda$ satisfies the CYBE (14.12) for every constant c .

Proof. Write $\lambda_{ij} = \lambda_i - \lambda_j$. The permutation operators satisfy $P_{12}P_{13} = P_{13}P_{23} = P_{23}P_{12}$ and $P_{13}P_{12} = P_{12}P_{23} = P_{23}P_{13}$ — both are the two ways of cycling three legs. Calling the difference of those two products K , all three commutators are $\pm K$:

$$[P_{12}, P_{13}] = K, \quad [P_{12}, P_{23}] = -K, \quad [P_{13}, P_{23}] = K.$$

So the left of (14.12) is

$$c^2 K \left(\frac{1}{\lambda_{12}\lambda_{13}} - \frac{1}{\lambda_{12}\lambda_{23}} + \frac{1}{\lambda_{13}\lambda_{23}} \right) = c^2 K \frac{\lambda_{23} - \lambda_{13} + \lambda_{12}}{\lambda_{12}\lambda_{13}\lambda_{23}} = 0,$$

because $\lambda_{12} + \lambda_{23} = \lambda_{13}$ identically. ■

The cancellation is a statement about the three spectral parameters and nothing else, which is why c is free: the normalisation of r is fixed by the Lax matrix it is paired with, not by the CYBE. [Lecture 9](#) fixes it for the periodic Toda chain, $c = -1$ in (9.10); the open chain of [Lecture 9](#)'s readings needs a different, constant r .

Two further families solve (14.12): the *trigonometric* r -matrices, in which $1/\lambda$ is replaced by $\cot \lambda$ or $\coth \lambda$ and the permutation is deformed, and the *elliptic* ones, built from Weierstrass functions. Belavin and Drinfeld classified the solutions *under hypotheses this course has not set up*: for r valued in $\mathfrak{g} \otimes \mathfrak{g}$ with $\mathfrak{g}$ a finite-dimensional *simple* Lie algebra, in difference form, and non-degenerate in their technical sense (the residue at the simple pole is proportional to the Casimir of an invariant form), every solution is equivalent to a rational, trigonometric or elliptic one *· cited*, and the three degenerate into one another in limits. Those hypotheses are doing real work — dropped, the statement is false, and already fails for $d = 1$, where every commutator in (14.12) vanishes and any function of λ with a simple pole at the origin solves it. The same trichotomy reappears in the Calogero–Moser potentials (optional lecture B) and in the quantum chains XXX, XXZ and XYZ — it is the same list every time, which is not a coincidence.

Reading

The open chain's r is not a solution

The open chain's constant r of (9.22) (Lecture 9's readings) is skew, $PrP = -r$, which is (9.12) for a constant r ; so Theorem 14.7 applies to it. Does it solve the classical Yang–Baxter equation (14.12)? **No.** Its left-hand side is

$$[r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}] = -\frac{1}{4} [P_{12}, P_{13}] \neq 0. \quad (14.13)$$

This identity is *cited*, not proved here: Babelon–Bernard–Talon, eq. (4.32), state it in operator form, and `scripts/check-l9-algebra.py` checks it symbolically for $N = 2, 3$. An equation of this shape, with a fixed non-zero right-hand side in place of the zero, is the *modified* CYBE named above.

And yet the Jacobi identity holds, as it must: the bracket is the canonical one on $\mathbb{R}^{2N}$. Theorem 14.7 asks only that the left of (14.13) commute with $L_1 + L_2 + L_3$, and it does, for *every* matrix L : P_{12} commutes with $L_1 + L_2$ (it only swaps them) and with L_3 (another leg), so it commutes with the sum, and so does P_{13} , and so does their commutator. This is Remark 14.9 in a real model: the CYBE is sufficient for Jacobi, and the Toda chain satisfies Jacobi without it.

How unique is r ? Adding $x\mathbb{1} + yP$ changes nothing in (9.21), since both commute with $L_1 + L_2$ (P only swaps the two legs). Babelon–Bernard–Talon, §4.7, derive (9.22) rather than check it, and show it is the only skew constant solution; `scripts/check-rmatrix.py` (V8) finds, at sampled points for $N = 3$, no constant solutions beyond $r + x\mathbb{1} + yP$. Of these, the skew one is r itself, which fails the CYBE, while $r \pm \frac{1}{2}P$ *do* solve the CYBE (Babelon–Bernard–Talon, (4.33); checked for $N = 2, 3$) and are not skew, so Theorem 14.7 as proved does not cover them. So no constant r for the open chain is both skew and a solution of the CYBE: the route “skew plus CYBE, hence Jacobi” is closed here, and Jacobi holds by the weaker condition above. That is not a defect of the model; it is why Lie-bialgebra theory has a modified equation.

Reading

14.11 Reordering three monodromies

Three auxiliary spaces 1, 2, 3 at parameters $\lambda_1, \lambda_2, \lambda_3$ give three relations (14.6), one for each pair, with $R_{ij} = R_{ij}(\lambda_i - \lambda_j)$.

Proposition 14.11 (Reordering three monodromies). *Let T satisfy (14.6) with an invertible R . Reversing $T_1T_2T_3$ one neighbouring pair at a time gives, along the two possible routes,*

$$T_1T_2T_3 = W_A (T_3T_2T_1) W_A^{-1} = W_B (T_3T_2T_1) W_B^{-1}, \quad W_A = R_{12}R_{13}R_{23}, \quad W_B = R_{23}R_{13}R_{12}.$$

Hence $W_B^{-1}W_A$ commutes with every $T_3T_2T_1$.

Proof. Write (14.6) as $T_iT_j = R_{ij}T_jT_iR_{ij}^{-1}$ for $i < j$. Route A exchanges the pairs 12, then 13, then 23: $T_1T_2T_3 = R_{12}T_2T_1T_3R_{12}^{-1} = R_{12}R_{13}T_2T_3T_1R_{13}^{-1}R_{12}^{-1} = W_A T_3T_2T_1 W_A^{-1}$, each R_{ij} commuting with the monodromy it does not touch. Route B exchanges 23, 13, 12 and gives W_B in the same way. Equating the two results gives the statement. ■

For a model that satisfies *RTT* this is automatic — the two routes rewrite one operator. It matters when *RTT* is taken as the *definition* of an algebra generated by the entries of T , as a Sklyanin bracket can be posited ([Lecture 9](#)'s readings): then the two routes yield the commutation condition of the proposition among the generators, and $W_A = W_B$ makes it automatic. That is sufficient, not necessary: for $R = \sigma^x \otimes \sigma^z$, which fails the Yang–Baxter equation, $W_B = -W_A$ and the condition is again automatic (the check script). It is [Remark 14.9](#) quantised: there the Jacobi identity required a combination of r 's only to commute with $L_1 + L_2 + L_3$, here the associativity of the products requires $W_B^{-1}W_A$ only to commute with the triple products, and for a particular model that is a weaker demand. What makes the Yang–Baxter equation the right thing to impose is, again, that it mentions no model. (The recipe $L_{an}(\lambda) = -R_{an}(-\lambda)$ of (14.7) used $R_{12} = R_{21}$; for a solution without that symmetry it is $L_{an}(\lambda) = -R_{na}(-\lambda)$.) For the Heisenberg chain nothing is left to choose: the check script finds that (14.4) fixes R up to a scalar for one and two sites, at generic λ, μ .

Reading

14.12 The Yang–Baxter equation at order $\hbar^2$, and more

YBE $\rightarrow$ *CYBE*. Give each line i a parameter λ_i and write $R_{ij} = R_{ij}(\lambda_i - \lambda_j)$, so (14.8) is $R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}$. Suppose $R = \mathbb{1} + i\hbar r + \hbar^2 s + O(\hbar^3)$, in difference form. The terms of order $\hbar$ are $r_{12} + r_{13} + r_{23}$ on both sides. Those of order $\hbar^2$ containing s are also the same on both sides; the rest are $(i\hbar)^2$ times the products of two r 's in the order of the factors, so the difference of the two sides at order $\hbar^2$ is

$$-\hbar^2 \left([r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}] \right) :$$

a solution of the Yang–Baxter equation has a classical r that solves the CYBE (14.12), in the per-leg form of § 14.10. For $R = \lambda - i\hbar P$, normalised by $1/\lambda$ (which does not change (14.8), the scalars being the same three on both sides), $r = -P/\lambda$ is [Proposition 14.10](#) at $c = -1$. This is the picture [Figure 14.8](#) promised: the classical slide is the first correction to the quantum one · *proved; the check script repeats it with generic matrices.*

The quantum determinant. At $\lambda - \mu = i$, $R(i) = i(1 - P)$ is $2i$ times the projector onto the antisymmetric part of $\mathbb{C}^2 \otimes \mathbb{C}^2$, and *RTT* says that $T_a(\mu + i)T_b(\mu)$ maps the antisymmetric line to itself. The operator by which it multiplies that line (central; for XXX a number) is the *quantum determinant*,

$$\text{qdet } T(\mu) = A(\mu + i)D(\mu) - B(\mu + i)C(\mu),$$

the counterpart of $\det T = 1$ for Toda (9.3). It commutes with every entry of T (Korepin, Bogoliubov and Izergin, §VII.8, in their conventions). For the universal algebra defined by *RTT* on a normalised formal series $T(u) = \mathbb{1} + T^{(1)}u^{-1} + \dots$, the Yangian $Y(\mathfrak{gl}_2)$, the coefficients of the quantum determinant generate the centre (Molev, ch. 1) · *cited*; a spin chain gives a representation of it. For the Heisenberg chain it is the number $[\mu(\mu + 2i)]^N$ (the check script, $N \leq 3$). [Lecture 15](#) meets its two factors as the eigenvalues of A and D on the reference state.

Other spins. For a spin s at each site, $L_{an}(\lambda) = \lambda + \frac{i}{2} + i\sigma_a \cdot \mathbf{S}_n$ satisfies (14.4) with the same R , by a verification that uses only the commutation relations of $\mathbf{S}$ · *cited: Korepin, Bogoliubov and Izergin, §VIII.4, (4.6)–(4.8), in their conventions; the check script verifies $s = 1$.* The transfer matrix still commutes with itself; but the regular point is lost, since $L(0)$ is

no longer a permutation, and a local Hamiltonian needs a transfer matrix with a spin- s auxiliary space (fusion: Kulish, Reshetikhin and Sklyanin 1981; not treated here).

Reading

14.13 Why the zero-curvature equation is not what is quantised

Lecture 8's and Lecture 9's readings derived the conservation of $t(\lambda)$ from the zero-curvature equation (8.23), $\dot{L}_n = V_{n+1}L_n - L_nV_n$, and Lecture 9 proved, separately, the Sklyanin bracket. This lecture quantised only the second. The zero-curvature equation is the equation of motion under one Hamiltonian, and in quantum mechanics equations of motion are not postulated: they follow from $[\cdot, H]$ once H is known. What has to be quantised is the *kinematics*, the brackets, and from them the commuting family and its Hamiltonians follow. The site-by-site transport survives as the monodromy (14.5), and the bracket as RTT . The Heisenberg equation for $T(\lambda)$ under H is then a consequence, and $[H, t(\lambda)] = 0$ is Theorem 14.3, not an input. That is the reason the classical limit of § 14.6 was taken in the bracket, not in the equation of motion.

Reading

14.14 The Toda pair, quantised

Lecture 12 set the Toda displacement beside Lecture 11's θ' as a counterpart. For the Toda pair itself the link can be checked. In Proposition 12.1 the relative coordinate $x = q_1 - q_2$ and its momentum $p = \frac{1}{2}(p_1 - p_2)$ have $H_{\text{rel}} = p^2 + e^x$, and the relative speed is $v = 2p$. Quantised,

$$-\hbar^2\psi'' + e^x\psi = p^2\psi.$$

With $\hbar = 1$ and $k = p$, the solution that decays under the wall is $\psi = K_{2ik}(2e^{x/2})$, the modified Bessel function: the chain rule turns this equation into Bessel's. Far from the wall, $x \rightarrow -\infty$, the small-argument form of K_ν gives · cited: DLMF §§10.25–10.27

$$\psi \approx \frac{1}{2} [\Gamma(-2ik) e^{ikx} + \Gamma(2ik) e^{-ikx}] :$$

an incoming wave e^{ikx} , reflected with the phase $\Gamma(2ik)/\Gamma(-2ik)$. Since $\Gamma(-2ik) = \overline{\Gamma(2ik)}$, the stationary-phase argument of § 11.3 puts the incoming packet on $x = 2kt + x_w$ and the outgoing one on $x = -2kt + x_w$, with $x_w = 2 \operatorname{Re} \psi_0(2ik)$ and $\psi_0 = \Gamma'/\Gamma$ the digamma function: the packet moves as if reflected by a hard wall at x_w . In the language of § 11.3, with $k = p_1 - p_2 = 2p$ the momentum difference, the reflected wave's phase is $\theta(k) = 2 \arg \Gamma(ik)$ up to a constant, and $\theta'(k) = 2 \operatorname{Re} \psi_0(ik) = x_w$: the wall is θ' . Restoring $\hbar$ shifts x by $2 \ln \hbar$ and replaces k by $p/\hbar$, so

$$x_w = 2 \ln \hbar + 2 \operatorname{Re} \psi_0(2ip/\hbar) \longrightarrow 2 \ln(2p) = 2 \ln v \quad (\hbar \rightarrow 0),$$

by Stirling's formula, $\operatorname{Re} \psi_0(iy) = \ln y + O(y^{-2})$. The classical orbit of Proposition 12.1, $x = \ln v^2 - v|t| + o(1)$, has its effective wall exactly there, and that wall is what displaces each momentum line by $\mp 2 \ln v$. So the classical displacement is the $\hbar \rightarrow 0$ limit of the quantum one (scripts/check-116-algebra.py checks the Bessel equation, the asymptotic

expansion and the effective-wall limit; the stationary-phase step is argued, not checked; Sutherland §4.2 does the same for the whole hyperbolic family) · *sketched*.

Summary.

- The Lax operator $L_{an}(\lambda) = \lambda + iP_{an} = -R_{an}(-\lambda)$ satisfies *RLL* with $R(\lambda) = \lambda - iP$ (Theorem 14.1), which is Lecture 13's Yang–Baxter equation.
- The train (Theorem 14.2): the monodromy satisfies *RTT*, and $[t(\lambda), t(\mu)] = 0$ (Theorem 14.3).
- The classical limit (Theorem 14.4 and Corollary 14.6): the quantum Toda chain satisfies *RLL* exactly with $R = \lambda - i\hbar P$, and at order $\hbar$ its *RTT* relation is Lecture 9's quadratic bracket with $r = -P/\lambda$.
- Next: Lecture 15 finds the Heisenberg Hamiltonian in $t(\lambda)$ and diagonalises the family by the algebraic Bethe ansatz.

Tutorial. Exercise Sheet 14.

References

E. K. Sklyanin, L. A. Takhtajan and L. D. Faddeev, *Theor. Math. Phys.* 40 (1979) 688 (the method); L. D. Faddeev, *How algebraic Bethe ansatz works for integrable model*, Les Houches lectures 1995, arXiv:hep-th/9605187, §§1–3; V. E. Korepin, N. M. Bogoliubov and A. G. Izergin, *Quantum Inverse Scattering Method and Correlation Functions*, Cambridge University Press (1993), §§VI.1–VI.5, §VII.8 (the quantum determinant), §VIII.4 (higher spin); M. C. Gutzwiller, *Ann. Phys.* 133 (1981) 304 and E. K. Sklyanin, *Lect. Notes Phys.* 226 (1985) 196 (the quantum Toda chain); A. Molev, *Yangians and Classical Lie Algebras*, American Mathematical Society (2007), Chapter 1 (the Yangian and its centre); O. Babelon, D. Bernard and M. Talon, *Introduction to Classical Integrable Systems*, Cambridge University Press (2003), Ch. 2–3 (the classical Yang–Baxter equation) and §4.7 (the open chain's r). **The conventions for the ordering of *RTT*, of the monodromy and for the sign in R differ between these sources; see § 14.2.**